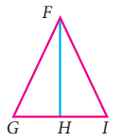
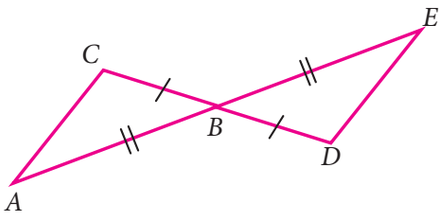
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9. In the given figure $FG = FI$ and H is midpoint of GI, prove that $\triangle FGH \cong \triangle FHI$

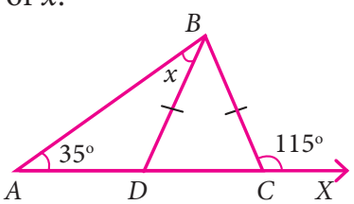


10. Using the given figure, prove that the triangles are congruent. Can you conclude that AC is parallel to DE.

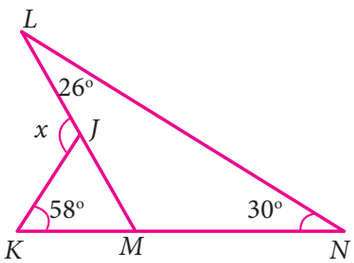


Challenge Problems

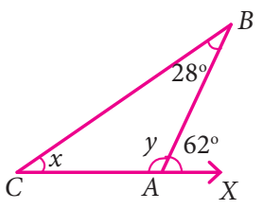
11. In given figure $BD = BC$, find the value of x.



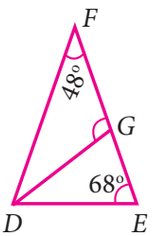
12. In the given figure find the value of x.



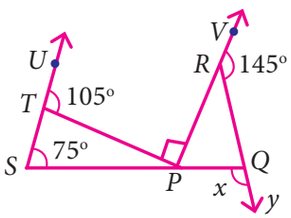
13. In the given figure find the values of x and y.



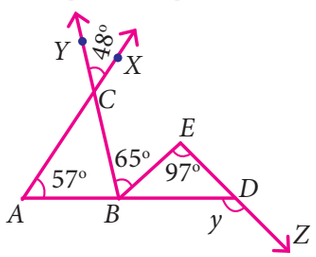
14. In ΔDEF, $\angle F = 48°$, $\angle E = 68°$ and bisector of $\angle D$ meets FE at G. Find $\angle FGD$.



15. In the figure find the value of x.



16. From the given figure find the value of y



Summary

- The sum of three angles in a triangle is 180°.
- The exterior angle of a triangle is equal to the sum of the two interior opposite angles.
- The exterior angles of a triangle add up to 360°.
- Two lines segments are congruent if they have the same length.
- Two angles are congruent if the measures of the angles are equal.

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